

UNIVERSIDADE ESTADUAL DE MONTES CLAROS – UNIMONTES
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ATIVIDADE AVALIATIVA III

RAMON LOPES DE QUEIROZ

MONTES CLAROS – MG
JUNHO/2026

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Atividade avaliativa apresentada para atendimento de requisito parcial para aprovação na disciplina Geometria Analítica e Álgebra Linear do Curso de Graduação em Bacharelado em Sistemas de Informação – 2º período

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MONTES CLAROS – MG

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Atribuição 4.3 (3, 6, 9, 10, 12, 16, 19).

2. a. LI. Não são múltiplos.

b. $D = \begin{vmatrix} -3 & 0 & 4 & -3 & 0 \\ 5 & -1 & 2 & 5 & -1 \\ 1 & 1 & 3 & 1 & 1 \\ 9 & 0 & 30 \end{vmatrix}$ $D = 9 + 0 + 30 - (-6 - 4 + 0)$ LI
 $D = 39$ $39 \neq 0$

c. LI. Não são múltiplos.

d. LD. 4 Vetores em um espaço R^3 , onde no máximo de vetores possível LI são 3.

6. a. $V_1 = (-1, 2, 3)$ $V_2 = 2V_1$ Como 1 e 2 são colineares, o conjunto é LD.
 $V_2 = (2, -4, -6)$ (Os três estão no mesmo plano.)
 $V_3 = (-3, 6, 0)$

b. $V_1 = (2, -1, 4)$ $D = \begin{vmatrix} 2 & -1 & 4 & 2 & -1 \\ 4 & 2 & 3 & 4 & 2 \\ 2 & 7 & -6 & 2 & 7 \\ -2 & 6 & 56 \end{vmatrix}$ $D = -24 + 6 + 56 - (34 + 42 + 16)$
 $V_2 = (4, 2, 3)$ $D = 0$
 $V_3 = (2, 7, -6)$ LD. Estão no mesmo plano.

c. $V_1 = (4, 6, 8)$ $V_2 = 2V_1$ LD. Estão no mesmo plano.
 $V_2 = (2, 3, 4)$ $V_3 = -V_2$
 $V_3 = (-2, -3, -4)$

9. $D = \begin{vmatrix} \lambda - \frac{1}{8} & -\frac{1}{8} & -\frac{1}{8} & \lambda - \frac{1}{8} \\ -\frac{1}{8} & \lambda - \frac{1}{8} & -\frac{1}{8} & \lambda \\ -\frac{1}{8} & -\frac{1}{8} & \lambda - \frac{1}{8} & -\frac{1}{8} \\ \lambda - \frac{1}{8} & -\frac{1}{8} & -\frac{1}{8} & \lambda \end{vmatrix}$ $D = \left(\lambda^3 - \frac{1}{8} - \frac{1}{8} \right) - \left(\frac{1}{4} \lambda + \frac{1}{4} \lambda + \frac{1}{4} \lambda \right)$
 $D = 4\lambda^3 - 3\lambda - 1 = 0$ $\lambda = 1 \text{ ou } -\frac{1}{2}$

$\lambda^4 = 4 \cdot 1^3 - 3 \cdot 1 - 1 = 0$ $4\lambda^2 + 4\lambda + 1 = 0$ $\lambda = -\frac{1}{2}$
 1 e 4 são $\lambda = -\frac{1}{2}$ $(2\lambda + 1)^2$
 $4 \ 4 \ 1 \ 0$ $2\lambda + 1 = 0$

$$10. C_1 V_1 + C_2 V_2 + C_3 V_3 = 0$$

$$C_1 = 0$$

$$C_2 = 0$$

$$C_3 = 0$$

$$a_1 V_1 + a_2 V_2 = 0$$

$$a_1 = 0 \text{ e } a_2 = 0$$

$$K V_1 + 0 V_2 + 0 V_3 = 0$$

$$a_1 V_1 + a_2 V_2 + 0 V_3 = 0$$

$$K = 0$$

$$12. C_1 V_1 + C_2 V_2 + C_3 V_3 = 0$$

$$C_1 V_1 + C_2 V_2 + C_3 V_3 + 0 V_4 = 0$$

$$a_1 V_1 + a_2 V_2 + a_3 V_3 + a_4 V_4 = 0$$

$$(C_1 V_1 + C_2 V_2 + C_3 V_3) + 0 = 0$$

$$0 = 0$$

$$a_1 = C_1 \quad a_2 = C_2 \quad a_3 = C_3 \quad a_4 = 0$$

$$16. V_1 = U - V$$

$$C_1 (U - V) + C_2 (V - W) + C_3 (W - U) = 0$$

$$C_1 = 1$$

$$V_2 = V - W$$

$$1 (U - V) + 1 (V - W) + 1 (W - U) = 0$$

$$C_2 = 1$$

$$V_3 = W - U$$

$$0 + 0 + 0 = 0$$

$$C_3 = 1$$

$$0 = 0$$

19. a. Transladando, eles comecam do mesmo ponto e apontam para direções diferentes. Não formam outro eixo linear. Dão LI.

b. Formam um triângulo fechado no espaço, sendo coplanares. Dão LD.

Atividade 4.4 (2, 3, 7, 9, 16).

2-a. $\{(0, 1), (3, 0)\}$ $D = \begin{vmatrix} 2 & 1 \\ 3 & 0 \end{vmatrix} = 0 - 3 = -3 \neq 0$ LI

b. $\{(4, 1), (-7, -8)\}$ $D = \begin{vmatrix} 4 & 1 \\ -7 & -8 \end{vmatrix} = -32 + 7 = -25 \neq 0$ LI

c. $\{(0, 0), (1, 3)\}$ LD. Possui vetor nulo.

d. $\{(3, 9), (-4, -12)\}$ $\begin{matrix} 3 \rightarrow 9 \\ 4 \rightarrow 0 \end{matrix} > (1 \rightarrow 3)$ Não múltiplos escalares entre si. LD

3-a. $D = \begin{vmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 3 & 3 \end{vmatrix}$ Matriz Triangular $D = 1 \cdot 2 \cdot 3 = 6 \neq 0$ LI. É linear.

b. $D = \begin{vmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{vmatrix}$ $D = 120 + 6 - 32 - (-20 + 22 + 16) = 26 \neq 0$ LI. É linear.

c. $D = \begin{vmatrix} 9 & -3 & 1 \\ 4 & 1 & 1 \\ 0 & -7 & 1 \end{vmatrix}$ $D = 2 + 0 + 28 - (0 - 14 - 12) = 26 - 26 = 0$ LD. Não é linear.

d. $D = \begin{vmatrix} 1 & 6 & 4 \\ 2 & 4 & -1 \\ -1 & 2 & 5 \end{vmatrix}$ $D = 20 + 6 + 16 - (-16 - 2 + 60) = 42 - 42 = 0$ LD. Não é linear.

7-a. $w = c_1 u_1 + c_2 u_2$ $c_1 = 3$
 $(3, 7) = c_1(1, 0) + c_2(0, 1)$ $c_2 = -7$
 $(3, 7) = (c_1, c_2)$

b. $(1, 1) = c_1(2, -4) + c_2(3, 8)$ $[w]_S = \begin{pmatrix} 5 \\ 28 \end{pmatrix}$
 $(1, 1) = (2c_1 + 3c_2, -4c_1 + 8c_2)$

$$\begin{cases} 2c_1 + 3c_2 = 1 \\ -4c_1 + 8c_2 = 1 \end{cases} \Rightarrow \begin{cases} 4c_1 + 6c_2 = 2 \\ -4c_1 + 8c_2 = 1 \end{cases} \quad 2c_1 + 3\left(\frac{3}{14}\right) = 1$$

$$14c_2 = 3 \quad c_1 = \frac{5}{28}$$

$$c_2 = \frac{3}{14}$$

c. $(a, b) = c_1(1, 1) + c_2(0, 2)$
 $(a, b) = (c_1, c_1 + 2c_2)$

$$\begin{cases} c_1 = a \\ c_1 + 2c_2 = b \end{cases} \quad \begin{cases} a + 2c_2 = b \\ c_2 = \frac{b-a}{2} \end{cases} \quad [w]_S = \begin{pmatrix} a \\ \frac{b-a}{2} \end{pmatrix}$$

9-a. $(2, -1, 3) = c_1(1, 0, 0) + c_2(2, 2, 0) + c_3(3, 3, 3)$
 $(2, -1, 3) = (c_1 + 2c_2 + 3c_3, 2c_2 + 3c_3, 3c_3)$

$$\begin{cases} c_1 + 2c_2 + 3c_3 = 2 \\ 2c_2 + 3c_3 = -1 \\ 3c_3 = 3 \end{cases} \quad \begin{matrix} c_3 = 1 \\ 2c_2 + 3(1) = -1 \\ c_2 = -2 \end{matrix} \quad \begin{matrix} c_1 + 2(-2) + 3(1) = 2 \\ c_1 = 3 \end{matrix}$$

$$[V]_S = (3, -2, 1)$$

$$b. (5, -12, 3) = C_1(1, 2, 3) + C_2(-4, 5, 6) + C_3(7, 8, 9)$$

$$\begin{cases} C_1 - 4C_2 + 7C_3 = 5 \\ 2C_1 + 5C_2 - 8C_3 = -12 \\ 3C_1 + 6C_2 + 9C_3 = 3 \end{cases}$$

$$\begin{bmatrix} 1 & -4 & 7 & 5 \\ 2 & 5 & -8 & -12 \\ 3 & 6 & 9 & 3 \end{bmatrix} \xrightarrow{L_2 - 2L_1, L_3 - 3L_1} \begin{bmatrix} 1 & -4 & 7 & 5 \\ 0 & 13 & -22 & -22 \\ 0 & 18 & -12 & -12 \end{bmatrix}$$

$$\frac{923}{50} C_3 = \frac{923}{50}$$

$$C_3 = 1$$

$$\begin{bmatrix} 1 & -4 & 7 & 5 \\ 0 & 1 & -\frac{22}{13} & -\frac{22}{13} \\ 0 & 18 & -12 & -12 \end{bmatrix} \xrightarrow{L_3 - 18L_2} \begin{bmatrix} 1 & -4 & 7 & 5 \\ 0 & 1 & -\frac{22}{13} & -\frac{22}{13} \\ 0 & 0 & \frac{923}{50} & \frac{923}{50} \end{bmatrix}$$

$$C_2 = \frac{-22}{13} + \frac{22}{13} = 0$$

$$C_1 - 4(0) + 7(1) = 5 \Rightarrow C_1 = -2$$

$$[V]_S = (-2, 0, 1)$$

$$16. a. (x, y) = x'(1, 0) + y'(1, 1)$$

$$(x, y) = (x' + y', y')$$

$$\begin{cases} y' = y \\ x' = x - y \end{cases}$$

$$y' = 1 \quad x' = 1 - 1 = 0 \quad x'y' = (0, 1)$$

$$b. y' = 0 \quad x'y' = (1, 0)$$

$$x' = 1 - 0 = 1$$

$$c. y' = 1 \quad x'y' = (-1, 1)$$

$$x' = 0 - 1 = -1$$

$$d. y' = b \quad x'y' = (a - b, b)$$

$$x' = a - b$$

Atividade 4.5 (1, 2, 3, 4, 5, 6).

1.
$$\begin{cases} x_1 + x_2 - x_3 = 0 \\ -2x_1 - x_2 + 2x_3 = 0 \\ -x_1 + x_3 = 0 \end{cases}$$

$$\begin{bmatrix} 1 & 1 & -1 & 0 \\ -2 & -1 & 2 & 0 \\ -1 & 0 & 1 & 0 \end{bmatrix} \xrightarrow[L_3+L_1]{L_2+2L_1} \begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \xrightarrow{L_3-L_2}$$

$$(x_1, x_2, x_3) = t(1, 0, 1)$$

$$\begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Base = $\{(1, 0, 1)\}$ | Dimensão = 1

2.
$$\begin{cases} 3x_1 + x_2 + x_3 + x_4 = 0 \\ 5x_1 - x_2 + x_3 - x_4 = 0 \end{cases}$$

$$\begin{bmatrix} 3 & 1 & 1 & 1 & 0 \\ 5 & -1 & 1 & -1 & 0 \end{bmatrix} \xrightarrow{L_2+L_1} \begin{matrix} 8x_1 + 2x_3 = 0 \\ x_3 = -4x_1 \end{matrix}$$

$$x_1 = s$$

$$(x_1, x_2, x_3, x_4) = (s, s-t, -4s, t)$$

$$3x_1 + x_2 + (-4x_1) + x_4 = 0$$

$$x_2 = s-t$$

$$(x_1, x_2, x_3, x_4) = s(1, 1, -4, 0) + t(0, -1, 0, 1)$$

$$x_2 = x_1 - x_4$$

$$x_3 = -4s$$

$$x_4 = t$$

Base = $\{(1, 1, -4, 0), (0, -1, 0, 1)\}$ | Dimensão = 2

3.
$$\begin{cases} x_1 - 4x_2 + 3x_3 - x_4 = 0 \\ 2x_1 - 8x_2 + 6x_3 - 2x_4 = 0 \end{cases}$$

$$x_1 = 4x_2 - 3x_3 + x_4$$

$$x_2 = a$$

$$x_3 = t$$

$$x_4 = u$$

$$(x_1, x_2, x_3, x_4) = (4a - 3t + u, a, t, u)$$

$$= a(4, 1, 0, 0) + t(-3, 0, 1, 0) + u(1, 0, 0, 1)$$

Base = $\{(4, 1, 0, 0), (-3, 0, 1, 0), (1, 0, 0, 1)\}$

Dimensão = 3

4.
$$\begin{cases} x_1 - 3x_2 + x_3 = 0 \\ 2x_1 - 6x_2 + 2x_3 = 0 \\ 3x_1 - 9x_2 + 3x_3 = 0 \end{cases}$$

$$x_1 = 3x_2 - x_3$$

Base = $\{(3, 1, 0), (-1, 0, 1)\}$

Dimensão = 2

$$(x_1, x_2, x_3) = (3a - t, a, t)$$

$$= a(3, 1, 0) + t(-1, 0, 1)$$

5-

$$\begin{cases} 2x_1 + x_2 + 3x_3 = 0 \\ x_1 + 5x_3 = 0 \\ x_2 + x_3 = 0 \end{cases}$$

$$x_1 = -5x_3$$

$$x_2 = -x_3$$

$$-8x_3 = 0$$

$$x_3 = 0$$

$$x_1 = 0 \quad x_2 = 0$$

$$\text{Base} = (0, 0, 0)$$

$$\text{Dimensão} = 0$$

6-

$$\begin{cases} x + y + z = 0 \\ 3x + 2y - 2z = 0 \\ 4x + 3y - z = 0 \\ 6x + 5y + z = 0 \end{cases}$$

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 3 & 2 & -2 & 0 \\ 4 & 3 & -1 & 0 \\ 6 & 5 & 1 & 0 \end{bmatrix}$$

 $L_2 - 3L_1$
 $L_3 - 4L_1$
 $L_4 - 6L_1$

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & -1 & -5 & 0 \\ 0 & -1 & -5 & 0 \\ 0 & -1 & -5 & 0 \end{bmatrix}$$

$$x - 5z + z = 0 \quad -y - 5z = 0$$

$$x = 4z$$

$$y = -5z$$

$$(x, y, z) = (4t, -5t, t) \\ = t(4, -5, 1)$$

$$\text{Base} = \{(4, -5, 1)\}$$

$$\text{Dimensão} = 1$$

Atividade 4.6 (1, 2, 6, 8, 12) -

1-a- $u_1 = (1, 0), u_2 = (0, 1), w = (3, -7)$

$$(3, -7) = c_1(1, 0) + c_2(0, 1) \quad [w]_S = (3, -7)$$

$$(c_1, c_2) = (3, -7)$$

b-
$$\begin{cases} 2c_1 + 3c_2 = 1 \\ -4c_1 + 8c_2 = 1 \end{cases} \times 2 = \begin{cases} 4c_1 + 6c_2 = 2 \\ -4c_1 + 8c_2 = 1 \end{cases} \quad 2c_1 + 3 \cdot \left(\frac{2}{14}\right) = 1$$

$$14c_2 = 3 \quad c_1 = \frac{5}{28}$$

$$c_2 = \frac{3}{14}$$

c- $(a, b) = c_1(1, 1) + c_2(0, 2)$ $c_1 = a$ $a + 2c_2 = b$

$$(a, b) = (c_1, c_1 + 2c_2) \quad c_2 = \frac{b-a}{2}$$

2-a- $(2, -1, 3) = c_1(1, 0, 0) + c_2(2, 2, 0) + c_3(3, 3, 3)$

$$\begin{array}{llll} 3c_3 = 3 & 2c_2 + 3c_3 = 1 & c_1 + 2(-2) + 3 \cdot 1 = 2 & [V]_S = (3, -2, 1) \\ c_3 = 1 & c_2 = -2 & c_1 = 3 & \end{array}$$

b- $(5, -12, 3) = c_1(1, 2, 3) + c_2(-4, 5, 6) + c_3(7, -8, 9)$

$$\begin{cases} c_1 - 4c_2 + 7c_3 = 5 \\ 2c_1 + 5c_2 - 8c_3 = -12 \\ 3c_1 + 6c_2 + 9c_3 = 3 \end{cases} \quad \begin{array}{l} c_1 = 1 - 2c_2 - 3c_3 \\ (1 - 2c_2 - 3c_3) - 4c_2 + 7c_3 = 5 \\ -3c_2 + 2c_3 = 2 \\ 2(1 - 2c_2 - 3c_3) + 5c_2 - 8c_3 = -12 \\ c_2 = 14c_3 - 14 \end{array}$$

$$\begin{array}{lll} -3(14c_3 - 14) + 2c_3 = 2 & c_2 = 14 \cdot 1 - 14 = 0 & c_3 = 1 - 2 \cdot 0 - 3(1) = -2 \\ c_3 = 1 & [V]_S = (2, 0, 1) & \end{array}$$

$$6-a. [u_1]_B = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$P_{B \leftarrow B'} = \begin{bmatrix} 2 & -3 \\ 1 & 4 \end{bmatrix}$$

$$[u_2]_B = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$$

$$b. P_{B' \leftarrow B} = (P_{B \leftarrow B'})^{-1}$$

$$P_{B' \leftarrow B} = \frac{1}{11} \begin{bmatrix} 4 & 3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 4/11 & 3/11 \\ -1/11 & 2/11 \end{bmatrix}$$

$$D_{P_{B \leftarrow B'}} = 2 \cdot 4 - (-3 \cdot 1) = 11$$

$$c. [w]_B = \begin{bmatrix} 3 \\ -5 \end{bmatrix} \quad [w]_{B'} = P_{B' \leftarrow B} \cdot [w]_B = \begin{bmatrix} 4/11 & 3/11 \\ -1/11 & 2/11 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ -5 \end{bmatrix} = \begin{bmatrix} -3/11 \\ -13/11 \end{bmatrix}$$

$$d. w = c_1 u_1 + c_2 u_2$$

$$2c_1 - 3c_2 = 3$$

$$\begin{bmatrix} 3 \\ -5 \end{bmatrix} = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -3 \\ 4 \end{bmatrix}$$

$$c_1 + 4c_2 = -5 \Rightarrow c_1 = -5 - 4c_2$$

$$2(-5 - 4c_2) - 3c_2 = 3$$

$$c_1 = -5 - 5 \left(\frac{-13}{11} \right)$$

$$c_2 = -\frac{13}{11}$$

$$c_1 = -3/11$$

$$[w]_{B'} = \begin{bmatrix} -3/11 \\ -13/11 \end{bmatrix}$$

$$8-a. \left[\begin{array}{ccc|ccc} -6 & -2 & -2 & -3 & -3 & 1 \\ -6 & -6 & -3 & 0 & 2 & 6 \\ 0 & 4 & 9 & -3 & -1 & -1 \end{array} \right] \xrightarrow{L_2 - L_1} \left[\begin{array}{ccc|ccc} -6 & -2 & -2 & -3 & -3 & 1 \\ 0 & -4 & -1 & 3 & 5 & 5 \\ 0 & 4 & 9 & -3 & -1 & -1 \end{array} \right] \xrightarrow{L_3 + L_2} \left[\begin{array}{ccc|ccc} -6 & -2 & -2 & -3 & -3 & 1 \\ 0 & -4 & -1 & 3 & 5 & 5 \\ 0 & 0 & 8 & 0 & 4 & 4 \end{array} \right]$$

$$\xrightarrow{\frac{1}{8}L_3} \left[\begin{array}{ccc|ccc} -6 & -2 & -2 & -3 & -3 & 1 \\ 0 & -4 & -1 & 3 & 5 & 5 \\ 0 & 0 & 1 & 0 & 1/2 & 1/2 \end{array} \right] \xrightarrow{L_2 + L_3} \left[\begin{array}{ccc|ccc} -6 & -2 & -2 & -3 & -3 & 1 \\ 0 & -4 & 0 & 3 & 11/2 & 7/2 \\ 0 & 0 & 1 & 0 & 1/2 & 1/2 \end{array} \right] \xrightarrow{L_1 + 2L_3} \left[\begin{array}{ccc|ccc} -6 & -2 & 0 & -3 & -5/2 & 2 \\ 0 & -4 & 0 & 3 & 11/2 & 7/2 \\ 0 & 0 & 1 & 0 & 1/2 & 1/2 \end{array} \right] \xrightarrow{-\frac{1}{4}L_2} \left[\begin{array}{ccc|ccc} -6 & -2 & 0 & -3 & -5/2 & 2 \\ 0 & 1 & 0 & -3/4 & -11/8 & -7/8 \\ 0 & 0 & 1 & 0 & 1/2 & 1/2 \end{array} \right]$$

$$\xrightarrow{L_1 + 2L_2} \left[\begin{array}{ccc|ccc} -6 & 0 & 0 & -9/2 & -11/4 & 3/4 \\ 0 & 1 & 0 & -3/4 & -11/8 & -7/8 \\ 0 & 0 & 1 & 0 & 1/2 & 1/2 \end{array} \right] \xrightarrow{-\frac{1}{6}L_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3/4 & 11/24 & -1/8 \\ 0 & 1 & 0 & -3/4 & -11/8 & -7/8 \\ 0 & 0 & 1 & 0 & 1/2 & 1/2 \end{array} \right]$$

$$b. \begin{bmatrix} -3 & -3 & 1 & -5 \\ 0 & 2 & 6 & 8 \\ -3 & -1 & -1 & -5 \end{bmatrix} \xrightarrow{L_3 - L_1} \begin{bmatrix} -3 & -3 & 1 & -5 \\ 0 & 2 & 6 & 8 \\ 0 & 2 & -2 & 0 \end{bmatrix} \xrightarrow{L_3 - L_2} \begin{bmatrix} -3 & -3 & 1 & -5 \\ 0 & 2 & 6 & 8 \\ 0 & 0 & -8 & -8 \end{bmatrix} \quad \begin{array}{l} -2C_2 = 8 \\ C_2 = 1 \end{array}$$

$$\begin{array}{l} 2C_2 + 6(1) = 8 \\ C_2 = 1 \end{array} \quad \begin{array}{l} -3C_1 - 3(1) + 1 = -5 \\ C_1 = 1 \end{array} \quad [w]_B = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 3/4 & 3/4 & 1/12 \\ -3/4 & -13/12 & -13/18 \\ 0 & 2/3 & 2/3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad [w]_B = P_B^{-1} \cdot a \cdot [w]_B$$

$$[w]_{B'} = \begin{bmatrix} 19/12 \\ -43/12 \\ 4/3 \end{bmatrix}$$

$$c. \begin{bmatrix} -6 & -2 & -2 & -5 \\ -6 & -6 & -3 & 8 \\ 0 & 4 & 9 & -5 \end{bmatrix} \xrightarrow{L_2 - L_1} \begin{bmatrix} -6 & -2 & -2 & -5 \\ 0 & -4 & -1 & 13 \\ 0 & 4 & 9 & -5 \end{bmatrix} \xrightarrow{L_3 + L_2} \begin{bmatrix} -6 & -2 & -2 & -5 \\ 0 & -4 & -1 & 13 \\ 0 & 0 & 8 & 8 \end{bmatrix} \quad \begin{array}{l} 6K_3 = 8 \\ K_3 = 8/6 \end{array}$$

$$\begin{array}{l} -4K_2 - \frac{4}{3} = 13 \\ K_2 = -43/12 \end{array} \quad \begin{array}{l} -6K_1 - 2 \cdot \left(\frac{-43}{12} \right) - 2 \cdot \left(\frac{4}{3} \right) = -5 \\ K_1 = 19/12 \end{array} \quad [w]_{B'} = \begin{bmatrix} 19/12 \\ -43/12 \\ 4/3 \end{bmatrix}$$

$$12. a. P_{B \rightarrow S} = \begin{bmatrix} 2 & -3 \\ 1 & 4 \end{bmatrix}$$

$$d. \begin{bmatrix} 5 \\ -3 \end{bmatrix} = C_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + C_2 \begin{bmatrix} -3 \\ 4 \end{bmatrix} \quad \begin{array}{l} 2C_1 - 3C_2 = 5 \\ C_1 = -3 - 4C_2 \end{array}$$

$$b. P_{S \rightarrow B} = \frac{1}{2 \cdot 4 - (-3) \cdot 1} \begin{bmatrix} 4 & 3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 4/11 & 3/11 \\ -1/11 & 2/11 \end{bmatrix}$$

$$2(-3 - 4C_2) - 3C_2 = 5 \quad C_1 = 1 \quad [w]_B = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$c. P_{B \rightarrow S} \cdot P_{S \rightarrow B} = \begin{bmatrix} 2 & -3 \\ 1 & 4 \end{bmatrix} \cdot \frac{1}{11} \begin{bmatrix} 4 & 3 \\ -1 & 2 \end{bmatrix}$$

$$e. w_S = \begin{bmatrix} 2 & -3 \\ 1 & 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}$$

$$d. \frac{1}{11} \begin{bmatrix} 11 & 0 \\ 0 & 11 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$e. w_B = \begin{bmatrix} 4/11 & 3/11 \\ -1/11 & 2/11 \end{bmatrix} \cdot \begin{bmatrix} 5 \\ -3 \end{bmatrix} = \begin{bmatrix} -3 \\ 11 \\ -13 \\ 11 \end{bmatrix}$$

Atividade 4.9 (6, 7, 8, 9, 12).

6-a- Sim. Ambos são combinação linear.

b- Sim. É uma transformação de rotação de 90° .

c- Sim. Ambos são lineares.

d- Não. x^2 não é linear.

e- Não. +1 somada sozinho quebra a linearidade.

7-a- Sim. É a transformação nula.

b- Não. Sempre retorna constantes não nulas.

c- Sim. Todos são lineares.

d- Não. y^2 não é linear.

e- Não. -1 impede que ela seja matricial.

8-a- $A = \begin{bmatrix} 2 & -3 & 0 & 1 \\ 3 & 5 & 0 & -1 \end{bmatrix}$

d- $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$

b- $A = \begin{bmatrix} 7 & 2 & -8 \\ 0 & -1 & 5 \\ 4 & 7 & -1 \end{bmatrix}$

c- $A = \begin{bmatrix} -1 & 1 \\ 3 & -2 \\ 5 & -7 \end{bmatrix}$

9-
$$\begin{cases} w_1 = 3x_1 + 5x_2 - 1x_3 \\ w_2 = 4x_1 - 1x_2 + 1x_3 \\ w_3 = 3x_1 + 2x_2 - 1x_3 \end{cases}$$

$$A = \begin{bmatrix} 3 & 5 & -1 \\ 4 & -1 & 1 \\ 3 & 2 & -1 \end{bmatrix}$$

12-a. $T(x) = [T] \cdot x$

$$[T] = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot x = \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 3-4 \\ 9-8 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad T(x) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

b. $T(x) = \begin{bmatrix} -1 & 2 & 0 \\ 3 & 1 & 5 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1+2+0 \\ -3+1+15 \end{bmatrix} = \begin{bmatrix} 3 \\ 13 \end{bmatrix}$

c. $T(x) = \begin{bmatrix} -2 & 1 & 4 \\ 3 & 5 & 7 \\ 6 & 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2x_1 + x_2 + 4x_3 \\ 3x_1 + 5x_2 + 7x_3 \\ 6x_1 - x_3 \end{bmatrix}$

d. $T(x) = \begin{bmatrix} -1 & 1 \\ 2 & 4 \\ 7 & 8 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_1 + x_2 \\ 2x_1 + 4x_2 \\ 7x_1 + 8x_2 \end{bmatrix}$

Atividade 5.1.

Autovetores

9.

$\lambda = 5$

$$1- Ax = \begin{bmatrix} 4 & 0 & 1 \\ 2 & 3 & 2 \\ 1 & 0 & 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 & 1 \\ 2 & 6 & 2 \\ 1 & 0 & 4 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} 5 \\ 10 \\ 5 \end{bmatrix} = 5 \cdot \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = 5x$$

Autovetores

$$2- Ax = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = 0 \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 0x$$

$\lambda = 0$

$$3 \text{ e } 4- a- \lambda^2 - \text{tr}(A)\lambda + \det(A) = 0$$

$$A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$$

$$\text{tr}(A) = 3 - 1 = 2$$

$$\det(A) = 3 \cdot (-1) - 0 \cdot 8 = -3$$

$$a- \text{Eq característica} - \lambda^2 - 2\lambda - 3 = 0$$

$$a- \text{Autovetores} = 3 \text{ e } -1$$

$$b- \begin{bmatrix} 10 & -9 \\ 4 & -2 \end{bmatrix}$$

$$\text{tr}(A) = 10 - 2 = 8$$

$$\det(A) = 10 \cdot (-2) - (-9 \cdot 4) = 16$$

$$b- \text{Eq. c} = \lambda^2 - 8\lambda + 16 = 0 \rightarrow (\lambda - 4)^2 = 0$$

$$b- \text{Autovetor} = \lambda = 4$$

$$c- \begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix}$$

$$\text{tr}(A) = 0$$

$$\det(A) = -12$$

$$c- \text{Eq} = \lambda^2 - 12 = 0$$

$$c- \text{Autovetor} = \lambda = \sqrt{12}$$

$$d- \begin{bmatrix} -2 & -7 \\ 1 & 2 \end{bmatrix}$$

$$\text{tr}(A) = 0$$

$$\det(A) = 3$$

$$d- \text{Eq. caract} = \lambda^2 + 3 = 0$$

$$d- \text{Autovetores} = \lambda = \sqrt{3}i \text{ e } -\sqrt{3}i$$

$$e- \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\text{tr}(A) = 0$$

$$\det(A) = 0$$

$$e- \text{Eq} = \lambda^2 = 0$$

$$e- \text{Autovetor} = 0$$

$$f- \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{tr}(A) = 2$$

$$\det(A) = 1$$

$$f- \text{Eq} = \lambda^2 - 2\lambda + 1 = 0 \rightarrow (\lambda - 1)^2 = 0$$

$$f- \text{Autovetor} = \lambda = 1$$

5. $(\lambda I - A)x = 0$

a. $\begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$ $\lambda_1 = 3$
 $\lambda_2 = -1$

$\lambda = 3 = \begin{bmatrix} 3-3 & 0 \\ -8 & 3+1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -8 & 4 \end{bmatrix}$ $x = t \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ Base de $E_3 = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$

$\lambda = -1 = \begin{bmatrix} -1-3 & 0 \\ -8 & -1+1 \end{bmatrix} = \begin{bmatrix} -4 & 0 \\ -8 & 0 \end{bmatrix}$ $x = \begin{bmatrix} 0 \\ t \end{bmatrix} = t \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $E_{-1} = \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$

b. $\begin{bmatrix} 10 & -9 \\ 4 & -2 \end{bmatrix}$

$\lambda = 4 = \begin{bmatrix} 4-10 & 9 \\ -4 & 4-(-2) \end{bmatrix} = \begin{bmatrix} -6 & 9 \\ -4 & 6 \end{bmatrix}$ $x = \begin{pmatrix} 3t \\ 2t \end{pmatrix} = t \begin{pmatrix} 3 \\ 2 \end{pmatrix}$ $E_4 = \left\{ \begin{bmatrix} 3 \\ 2 \end{bmatrix} \right\}$

c. $A = \begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix}$

$\lambda = 2\sqrt{3} = \begin{bmatrix} 2\sqrt{3} & -3 \\ -4 & 2\sqrt{3} \end{bmatrix}$ $x = t \begin{pmatrix} \sqrt{3} \\ 2 \end{pmatrix}$ $E_{2\sqrt{3}} = \left\{ \begin{pmatrix} \sqrt{3} \\ 2 \end{pmatrix} \right\}$

$\lambda = -2\sqrt{3} = \begin{bmatrix} -2\sqrt{3} & -3 \\ -4 & -2\sqrt{3} \end{bmatrix}$ $x = t \begin{pmatrix} \sqrt{3} \\ -2 \end{pmatrix}$ $E_{-2\sqrt{3}} = \left\{ \begin{pmatrix} \sqrt{3} \\ -2 \end{pmatrix} \right\}$

d. $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$\lambda = 0 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$x = \begin{bmatrix} 2 \\ t \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

e. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\lambda = 1 = \begin{bmatrix} 1-1 & 0 \\ 0 & 1-1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$E_1 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$

$$12.a. \begin{bmatrix} -1 & 6 \\ 0 & 5 \end{bmatrix} \quad \lambda_1 = -1 \\ \lambda_2 = 5$$

$$b. \begin{bmatrix} 3 & 0 & 0 \\ -2 & 7 & 0 \\ 9 & 8 & 1 \end{bmatrix} \quad \lambda_1 = 3 \\ \lambda_2 = 7 \\ \lambda_3 = 1$$

$$c. \begin{bmatrix} -1/3 & 0 & 0 & 0 \\ 0 & -1/3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1/2 \end{bmatrix} \quad \lambda_1 = -1/3 \\ \lambda_2 = -1/3 \\ \lambda_3 = 1 \\ \lambda_4 = 1/2$$

$$13. A = \begin{bmatrix} 1 & 3 & 7 & 11 \\ 0 & 1/2 & 3 & 8 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 2 \end{bmatrix} \quad \lambda_1 = 1 \\ \lambda_2 = 1/2 \\ \lambda_3 = 0 \\ \lambda_4 = 0$$

$$A^9 = \begin{bmatrix} 1^9 & & & \\ & (1/2)^9 & & \\ & & 0^9 & \\ & & & 0^9 \end{bmatrix} = \begin{bmatrix} 1 & & & \\ & 1/512 & & \\ & & 0 & \\ & & & 0 \end{bmatrix}$$

$$A^9 = \begin{pmatrix} 1, \frac{1}{512}, 0, 0 \end{pmatrix}$$

$$15.a. \begin{bmatrix} 4 & -1 \\ 2 & 1 \end{bmatrix} \quad \text{tr} = 5 \quad \det = 6 \quad \text{Eq} = \lambda^2 - 5\lambda + 6 = 0 \rightarrow (\lambda - 2)(\lambda - 3) = 0$$

$$\text{Autore} = \lambda_1 = 2 \quad \lambda_2 = 3$$

$$(2I - A)x = \begin{bmatrix} 2-4 & 1 \\ -2 & 2-1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ -2 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad y = 2x$$

$$(3I - A)x = \begin{bmatrix} 3-4 & 1 \\ -2 & 3-1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -2 & 2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad y = x \quad y = 2x \text{ e } y = x$$

$$b. \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \quad \text{tr} = 0 \quad \det = 1 \quad \lambda^2 + 1 = 0 \\ \lambda = \pm i$$

$$16.a. p(\lambda) = \lambda^3 - 2\lambda^2 + \lambda + 5$$

$$p(0) = 0^3 - 2 \cdot 0^2 + 0 + 5 = 5$$

$$\det A = \frac{5}{(-1)^3} = -5$$

$$b. p(\lambda) = \lambda^4 - \lambda^3 - 7$$

$$p(0) = 0^4 - 0^3 - 7 = -7$$

$$\det A = \frac{-7}{(-1)^4} = -7$$

$$17.a. p(\lambda) = \det(\lambda I - A) =$$

$$\begin{vmatrix} \lambda - a_{11} & -a_{12} & \dots & -a_{1n} \\ -a_{21} & \lambda - a_{22} & \dots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{m1} & -a_{m2} & \dots & \lambda - a_{nn} \end{vmatrix}$$

$$(\lambda - a_{11}) \cdot (\lambda - a_{22}) \cdot \dots \cdot (\lambda - a_{nn})$$

$$\lambda \cdot \lambda \cdot \lambda \cdot \dots \cdot \lambda = \lambda^n$$

$$b. (\lambda - a_{11})(\lambda - a_{22}) \cdot \dots \cdot (\lambda - a_{nn})$$

$$1 \cdot \lambda^n$$